Ariel A. Arévalo Alvarado

AI/ML Engineering Lead

"Built 2 enterprise AI platforms from scratch as founding/primary engineer, delivering 50+ production agents and a product concept validated by a $9M investment from OpenAI in a similar initiative."

Contact

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Work History

CRSS, Remote
Founding AI/ML Engineer — *Jun 2025 – Present*

- Designed an apprentice agent workflow platform (AutoActivity) from zero to production as the lead founding engineer, delivering a 4-repository system with 1,162 commits spanning backend (FastAPI), frontend (Playwright browser event trace capture), and Azure ML data pipelines.
- Architected and built across 5+ client-specific AI prototypes—chatbots, document classifiers, and data analysis agents—using LangChain, MCP, Azure AI Foundry, and Slack integrations.
- Established an AI engineering organization spanning 28 repositories and 1,860 commits, including production platforms, client prototypes, documentation hubs, and team training resources.

Skills: Python, FastAPI, Azure ML, LangChain, Docker, MCP, Alembic, Azure AI Foundry, System Architecture

Colibri Group, Remote
Senior AI Engineer — *Jul 2025 – Present*

- Designed a zero-deployment agent configuration architecture ("Instant Agents") that reduced per-agent deployment effort by 45% through a control plane/data plane separation with immutable versioning and automated component discovery. This allowed us to scale from 6 to 50+ agents at $8.52/agent/month with under 1% infrastructure overhead
- Built the Rubi enterprise AI platform from scratch as the primary developer (573 commits), implementing agent runtime orchestration, JWT security, persistent memory, and a self-serve agent builder on AWS Bedrock AgentCore.

Skills: Python, AWS Bedrock AgentCore, MCP, Abacus AI, System Architecture

Datasite, Heredia
AI/ML Engineer — *Apr 2024 – Jun 2025*

- Designed the first end-to-end MLOps pipeline at Datasite, replacing ad-hoc Jupyter notebook experimentation with reproducible Azure ML pipeline steps incorporating data versioning and experiment tracking.
- Architected a medallion data architecture (Bronze/Silver/Gold) for production ML workflows, establishing standardized ingestion, validation, and feature engineering stages on Airflow and Azure ML.
- Maintained and extended existing production systems running classical NLP methods (NER, PII detection, BM25 search), regression models with Scikit-learn, and deep learning classifiers (MLPs) using PyTorch and Keras.

Skills: Python, Pandas, Azure ML, Apache Airflow, MLOps, Scikit-learn, Keras, PyTorch, NLP, Deep Learning

TransUnion, Heredia
Data Engineer *Dec 2022 – Apr 2024*

- Led development of the initial ETL pipeline synthesizing IPv6 geolocation data for the industry-leading IP GeoPoint product, designing Spark SQL and Spark ML workflows on Google Cloud Dataproc.
- Maintained an ETL pipeline synthesizing IP reputation data to determine whether users were malicious and/or bots, making use of a Decision Tree model running in J2EE.

Skills: Scala, Java, Python, Apache Spark, Spring Framework, GCP Dataproc

Snap Finance, Alajuela
Software Developer *Dec 2021 – Dec 2022*

- Developed originations back-end services across monolith (J2EE), microservices (Spring), and serverless (AWS Lambda) architectures for consumer financing applications.

Skills: Java, JavaScript, Python, Spring Framework, J2EE, AWS Lambda

Publications In Progress

K. Villalobos-Solís, E. Víquez-Mora, A. Badilla-Olivas, E. Vílchez-Lizano, and A. Arévalo-Alvarado, "Multimodal Foundation Models for Zero-Shot User Activity Classification", *accepted to JOCICI*, 2026.

Professional Development

BlueDot Impact, AGI Strategy *Feb 2026 – Mar 2026*

AAAI Workshop, Towards Execution-Grounded Automated AI Research *Feb 2026*

Projects

Rigger – Declarative coding agent harness framework *Ongoing*

- Built a framework for declarative coding agent harness management, enabling users to define harnesses as code or configuration and managing the development lifecycle based on user workflow specifications.

Skills: Python, Claude Agent SDK, Automated Software Engineering, Evolutionary Algorithms

ClimateFact – AI-powered climate statement fact-checker *Spring 2025*

- Built an AI fact-checking system using LangGraph and Azure OpenAI to detect contradictions in climate statements against the IPCC AR6 report, leveraging DeBERTa natural language inference models for semantic validation.

Skills: Python, LangGraph, Azure OpenAI, NLP, NLI, DeBERTa

Education

BSc. University of Costa Rica, Computer Science *Dec 2026 (expected)*

BSc. University of Costa Rica, Mechanical Engineering *On hold since 2020*

Languages

English: Native **Spanish**: Native